CORRECTNESS-ADMISSIBLE PATHS AND REALIZED COSTS IN CONSUMER SPECULATIVE DECODING

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ABSTRACT

Speculative decoding saves time only when the execution path satisfies the required output contract and its realized costs fit within the progress it commits. We characterize four consumer systems under greedy token equivalence, separating target-forward microbenchmarks from complete round costs and isolated draft throughput from stateful draft execution. On MacBook Air M4, a representative workload is exact but slower, whereas a separately preregistered HumanEval workload achieves 1.387× speedup (95% CI [1.351, 1.422]) with 90/90 exact pairs. On RTX 5070 Ti Laptop and Arc B570, correctness excludes fast verification paths; serialization restores equality in the tested RTX controls but does not establish workload-global equality on B570. Core Ultra X7 358H CPU and NPU draft paths are exact yet unprofitable, with complete target-side work and full-prefix replay plus proposal dominating their respective costs. These observations identify correctness-admissible execution and the realized state interface as necessary measurement boundaries for this study, without asserting a hardware ranking or a measured runtime optimization.

Manuscript v0.1—hardware-complete / runtime-reserved. Title and abstract provisional.

1 INTRODUCTION

Speculative decoding uses a cheaper draft to propose tokens and a target model to validate them, amortizing target execution over multiple committed outputs (Leviathan et al., 2023). Its systems benefit depends on the work actually executed. A target backend may evaluate a candidate block cheaply yet produce a different greedy token from the sequential reference. A draft accelerator may decode efficiently in isolation yet repeatedly reconstruct state when integrated into a speculative loop. Both effects change which costs count toward a usable acceleration.

We ask: *when execution is filtered by a strict greedy token-equivalence contract, how do realized verification and state-interface costs determine whether speculation pays on the tested consumer systems?* We study MacBook Air M4, Core Ultra 9 275HX with RTX 5070 Ti Laptop, Core i5-13400 with Arc B570, and Core Ultra X7 358H with Arc B390 and CPU/NPU drafting. The comparison spans frozen runtime configurations rather than independently controlled hardware factors.

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The observations resist a platform-wide success or failure label. M4 is exact but unprofitable on a representative workload and exact and profitable on a separately preregistered code workload. The RTX target has cheap batched execution, but the frozen fast path violates token equivalence; exact serialized controls are slower. B570 serialization improves agreement without making the full workload exact. On 358H, direct round accounting reveals different bottlenecks for CPU and NPU drafting despite exact outputs on both.

Prior work already studies speculative-decoding performance, heterogeneous placement, cost/commit accounting, and output reproducibility (Zhang et al., 2025; Lv et al., 2025; Liu et al., 2026; Kim et al., 2026; Gond et al., 2026). Our contribution is a bounded empirical combination: correctness changes the usable execution path, while the actual state interface can erase an isolated draft-device advantage. We support this with frozen workloads, paired controls, explicit evidence classes, and direct stateful cost measurements where the runner exposes them.

Specifically, we provide a cross-runtime characterization that retains positive, negative, and correctness-blocked outcomes; an accounting discipline separating complete round costs from forward-only proxies; within-platform verification-path controls; and a direct heterogeneous state-interface decomposition. The M4 workload contrast bounds the positive finding instead of generalizing it. These hardware-study conclusions stand independently of any subsequent runtime intervention.

2 Correctness Contract and Realized-Cost Accounting

2.1 A path-qualified output contract

A greedy speculative round proposes a draft sequence, verifies it with the target, and commits the matching prefix followed by a correction or bonus token when the remaining output budget permits. Rejected candidate state must be removed or reconstructed consistently. Our contract requires generated token IDs to equal the matched target-only greedy output under the frozen configuration. It is an empirical property of a model, runtime, execution path, and workload, not a property inferred from the algorithm name.

We call a tested path *admissible* only within the workload scope for which that contract passes. A failed pair blocks a workload-global optimization claim even if the timing is favorable. A small qualification probe establishes eligibility for broader testing, not universal equivalence. We retain blocked-path timings as diagnostics with their correctness status visible. This deliberate contract is stricter than an approximate-quality deployment objective; HumanEval token equality does not establish functional correctness or pass@k.

2.2 Complete round costs

Let T_b be the matched target-only cost per output token. The realized draft time T_d^{real} denotes the realized draft-subsystem time per round, including any prefix replay performed by its interface. The complete target time $T_{\text{target,round}}^{\text{real}}$ includes all measured target-side round work: candidate evaluation and any correction, bonus, or rollback work inside that region. Residual synchronization, transfer, orchestration, and bookkeeping time not included in either region is T_x. Let C count *all* committed output tokens. With consistent timing and aggregation boundaries,

$$S = \frac{T_b C}{T_d^{\text{real}} + T_{\text{target,round}}^{\text{real}} + T_x} \tag{1}$$

$$= \frac{C}{D_{\text{real}} + R_{\text{target}} + X}, \tag{2}$$

where each denominator term is normalized by T_b. Break-even requires $C > D_{\text{real}} + R_{\text{target}} + X$. These identities organize measurements; they are not a new profitability theory. Cost/commit decompositions already appear in runtime-controlled studies (Kim et al., 2026).

At an aggregate level, round means and matched total-time/token accounting must use the same weights. A ratio of total times need not equal a geometric mean of prompt speedups. Likewise, accepted draft tokens per round omit correction or bonus output and cannot automatically be used as C. We do not infer unmeasured replay-only, transfer-only, or correction-only components by subtraction from incompatible experiments.

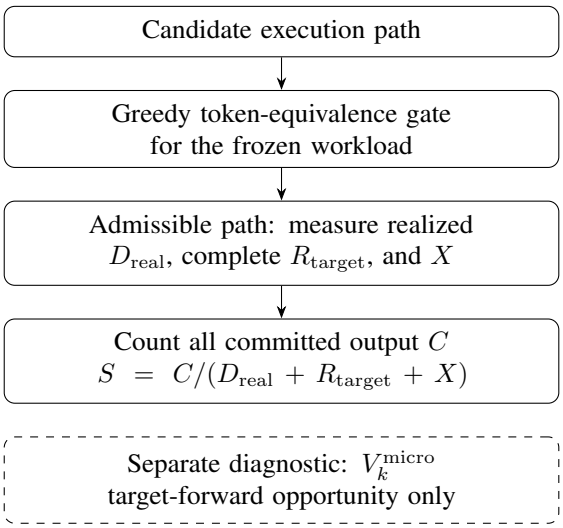


Figure 1. What must be measured after correctness filtering? Complete round accounting applies only to an admissible path. A failed gate leaves diagnostics; the dashed microbenchmark box is not substituted into the realized denominator.

2.3 A separate target-forward proxy

For context n and path e, define

$$V_k^{\text{micro}}(n, e) = \frac{T_{\text{target forward}}(k, n, e)}{T_{\text{target decode}}(1, n)}. \tag{3}$$

This microbenchmark asks whether a target can process a causal block near the cost of a single decode. It omits draft generation and can omit target correction/bonus and state-management work. Consequently V_k^{micro} is not automatically R_{target}. Figure 1 separates the opportunity measured by the proxy from the costs paid by a qualified speculative round.

3 Experimental Methodology and Evidence Classes

3.1 Frozen systems and workloads

Table 1 summarizes the completed campaigns. Final comparisons process one request at a time (sequence batch one). The final target is Qwen3-8B Q8_0; the native draft is Qwen3-0.6B Q8_0. The 358H heterogeneous adapter instead uses the qualified symmetric INT4 OpenVINO draft. We retain this quantization difference rather than describing the stacks as identical. Earlier M4 MLX-LM precision/correctness failures belong to qualification history and are distinct from the final llama.cpp Metal path.

The M4 representative sample freezes 24 previously unseen Spec-Bench prompts (Xia et al., 2024), with four from each

Table 1. Frozen platform and runtime scope. All final target models are Qwen3-8B Q8_0. All campaigns use greedy generated-token equivalence to a matched target-only reference. Hardware and software co-vary.

System	Runtime / target backend	Draft model / placement	Evidence scope
MacBook Air M4	llama.cpp b10809 / Metal	Qwen3-0.6B Q8_0 / Metal	Representative 24Q and separate HumanEval30
Core Ultra 9 275HX + RTX 5070 Ti Laptop	llama.cpp b10839 / CUDA	Qwen3-0.6B Q8_0 / CPU or same GPU	Blocked fast path; exact serialized controls
Core i5-13400 + Arc B570	llama.cpp b10839 / Vulkan0	Qwen3-0.6B Q8_0 / B570	Formal gate failure; later diagnostics
Core Ultra X7 358H + Arc B390	llama.cpp b10839 / Vulkan0; OpenVINO draft	Qwen3-0.6B symmetric INT4 / CPU or NPU	Closed Phase 3; direct stateful decomposition

of six workload groups and prior diagnostic IDs excluded. It uses three randomized paired repetitions. A separate preregistration freezes 30 HumanEval IDs (Chen et al., 2021) before prompt download or inspection; fixed $k = 2$ is the primary condition. The registered confidence-threshold conditions remain secondary. The subsequent HumanEval comparisons retain the frozen code-workload scope, three paired repetitions, greedy decoding, and a 128-token cap. They do not execute generated code.

Primary timing compares matched decode regions, excluding model loading and initial prompt prefill. In the native campaigns we use the frozen runtime decode counters; 358H explicitly times the complete post-prefill speculative loop. Replaying the committed prefix *inside* that loop remains charged to drafting. Thus the measurements are conditional decode costs, not whole-request latency including startup. Target-forward controls have their own narrower timing boundary. Different runtime versions, quantization, and device paths prevent a hardware-only causal comparison.

3.2 Evidence classes and uncertainty

Primary results answer a frozen workload question after its correctness gate. **Secondary** results are registered variants or explicitly bounded follow-up controls. **Mechanism-supported** claims use path controls or timed regions. **Diagnostic** values describe correctness-blocked or otherwise restricted paths. These labels describe permitted inference, not just measurement precision. In particular, the exact RTX serialized controls are secondary evidence; they do not replace its blocked primary fast path. B570 follow-ups remain diagnostic after the formal stop.

For each prompt we reduce its three paired baseline/speculative ratios, then report the geometric mean across prompts. M4 and RTX use a within-prompt geometric mean, as does the B570 diagnostic analysis; 358H uses a within-prompt median. Intervals are 95% prompt bootstraps. M4 HumanEval uses 100,000 resamples; the other reported campaigns use 10,000. Frozen seeds and endpoint conventions are given in the separate appendix. We preserve campaign-specific estimators and finite-bootstrap input order rather than harmonizing them after observing outcomes.

These intervals quantify prompt sampling conditional on the campaign. They do not encompass all session, temperature, or runtime-state uncertainty. Randomized adjacent pairing reduces drift but does not remove it. We therefore keep the representative and code-conditioned M4 studies separate, including their differing session conditions.

3.3 Provenance and energy boundaries

Frozen definitions, processed summaries, scripts, and archived evidence anchor every quantitative claim in an internal ledger. Windows raw per-run provenance is weaker than the other completed platforms, although committed pair tables and mechanism captures support recomputation. The 358H reconstructed draft source is `STRONGLY_SUPPORTED`; exact historical upstream identity is not proven. The header-only blocked-preflight derivative is excluded from final analysis.

Energy is interpreted only within its qualified sensor boundary. M4 measures package aggregate excluding separate DRAM; B570 measures adapter board energy. RTX board and CPU-package domains are not a common whole-system boundary, and its serialized HumanEval follow-up lacks an admissible formal energy comparison. No end-to-end Phase-3 energy claim is made for 358H. We do not rank platform energy efficiency.

4 Correctness Changes the Usable Execution Path

4.1 Cheap physical batching is not sufficient

The final M4 Metal path combines exact frozen-workload outputs with a target-forward proxy near one baseline step. Across its tested contexts, V_2^{micro} is approximately 1.02–

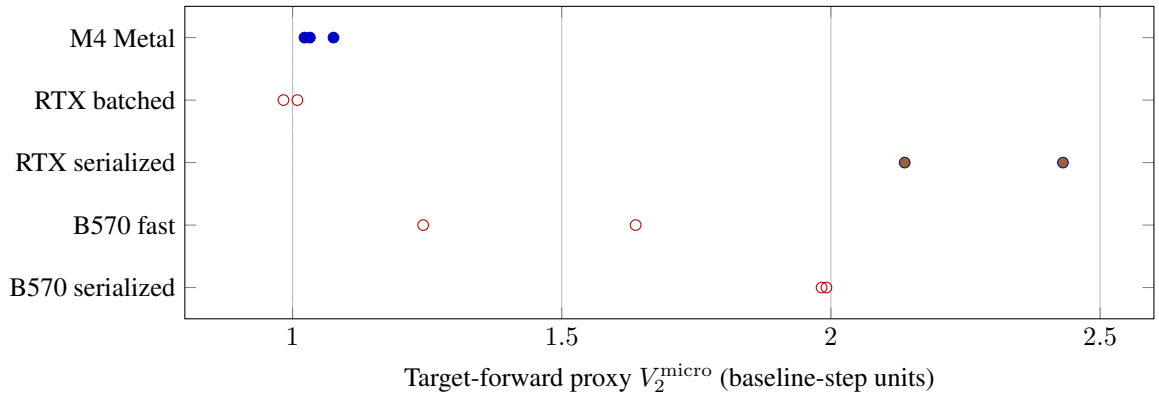


Figure 2. How does correctness change usable verification? Filled marks: paths exact on the frozen reported workloads; open marks: correctness-blocked paths (including B570 serialized). Each mark is one context cell; nearby cells can overlap. M4 contexts: 512/2048/8192; RTX 5070 Ti Laptop depths: 0/256; B570 contexts: 512/2048. Contexts, runtimes, and controls differ: this is not a hardware ranking. The plotted proxy is not complete R_{target}.

Table 2. Verification/admissibility mechanism. Proxy ranges span measured contexts, not confidence intervals. Full target-round R_{target} is not exposed by these microbenchmarks.

Path	V_2^{micro}	Qualification
M4 Metal	1.02–1.08	Frozen workloads exact
RTX batched	0.98–1.01	Correctness-blocked
RTX serial	2.14–2.43	Exact secondary control
B570 fast	1.24–1.64	45/90 exact; diagnostic
B570 serial	1.98–1.99	60/90 exact; diagnostic

1.08. This is a batching opportunity, not a complete round decomposition, and does not by itself explain whether a workload is profitable.

On RTX 5070 Ti Laptop, the batched CUDA target also has near-one-step two-token forward cost: $V_2^{\text{micro}} = 1.009$ and 0.983 at the two measured depths. The intended fast HumanEval execution nevertheless fails the frozen token-equivalence gate. Its timings cannot support an admissible optimization claim. The serialized control changes the micro-batch to one and raises the measured proxy to 2.431 and 2.137; the full frozen exact-control workload passes. Figure 2 therefore relates a physical opportunity to its empirically qualified status, rather than ranking GPUs.

4.2 A bounded numerical divergence

The frozen CUDA reproducer repeats the HumanEval/106 divergence in three of three runs at the same captured location. Sequential and serialized execution select token 2790, whereas batched execution selects 2629. A separate same-commit diagnostic build captures the corresponding score-margin change from $+0.080948$ to -0.126457. That numerical instrumentation does not replace the original formal binary or its measurements.

The evidence supports stable batch-shape-dependent greedy argmax divergence in this llama.cpp/CUDA/Q8_0 configuration. It does not identify a universal CUDA failure or a specific responsible kernel. Floating-point and batch-dependent output differences are established research topics (Yuan et al., 2025; Gond et al., 2026); the systems consequence here is that the cheap path is removed from the set admitted by the study's contract.

4.3 Serialization does not establish exactness

B570 provides a complementary boundary. Its formal v2 correctness gate fails, so the preregistered stop rule prevents a formal verification/performance claim from that execution. Later fast-path diagnostics yield 45/90 exact pairs. The unique serialized fallback improves this to 60/90, with 20/30 wholly exact prompts, but still fails workload-global equality. Its V_2^{micro} is approximately 1.983–1.992 and remains diagnostic.

A smaller serialized probe had passed before the full workload exposed mismatches. Passing such a gate is evidence for its tested scope, not proof of broader correctness. We therefore use "serialized V_2^{micro}" or "V2_serialized" for this B570 path. An execution property must not be renamed as an exactness guarantee.

Table 3. Headline decode outcomes. Speedup is baseline/speculative time, aggregated by the campaign's frozen prompt estimator. Brackets are 95% prompt-bootstrap CIs. Exact pairs and faster prompts are separate metrics. B570 timing is diagnostic, despite a reported CI.

Path / workload	Speedup [95% CI]	Exact pairs	Faster prompts	Class
M4 representative 24Q	0.842 [0.809, 0.877]	72/72	1/24	Primary
M4 HumanEval fixed-$k = 2$	1.387 [1.351, 1.422]	90/90	30/30	Primary
CPU to RTX 5070 Ti Laptop	0.404 [0.390, 0.418]	90/90	0/30	Second.
RTX 5070 Ti Laptop to itself	0.650 [0.629, 0.673]	90/90	0/30	Second.
B570 serialized fallback	0.3087 [0.2927, 0.3246]	60/90	0/30	Diag.
358H CPU draft	0.734 [0.693, 0.771]	90/90	0/30	Primary
358H NPU draft	0.352 [0.333, 0.369]	90/90	0/30	Primary

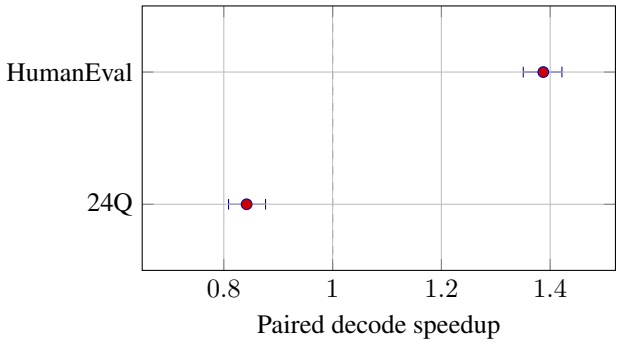


Figure 3. Is the M4 positive result representative? The frozen 24Q result is negative (72/72 exact); separately preregistered HumanEval is positive (90/90 exact). Bars are campaign-specific 95% prompt-bootstrap intervals. Both use the qualified Metal path.

5 Hardware Results by Regime

5.1 M4: workload-conditioned profitability

The representative frozen M4 sample produces 0.842× speedup (95% CI [0.809, 0.877]), with 72/72 exact pairs and only 1/24 prompts faster. The separately preregistered HumanEval fixed-$k = 2$ study produces 1.387× [1.351, 1.422], with 90/90 exact pairs and 30/30 prompts faster. Figure 3 retains both; the positive code-conditioned result does not replace or pool with the negative representative result.

This establishes existence of an exact profitable workload regime on the measured M4 stack. It does not establish general M4 profitability or isolate unified memory as its cause. The workloads and session conditions differ, and the earlier discovery effect size is not promoted as a confirmatory estimate. Historical MLX failures also do not negate the final qualified Q8_0 Metal result.

For primary fixed-$k = 2$ HumanEval, the package-efficiency ratio (baseline/speculative J per output token) is 1.013× [0.990, 1.034]. The interval includes one: energy is neutral within uncertainty. The registered $p_{\min} = 0.9$ condition has secondary energy evidence, discussed separately in the appendix. The preregistered corrected synthetic ceiling was not executed. We retain that deviation rather than treating the earlier synthetic diagnostic as its replacement.

5.2 RTX 5070 Ti Laptop: exact controls remain negative

The serialized CPU-to-RTX control reaches 0.404× [0.390, 0.418], while the same-GPU draft control reaches 0.650× [0.629, 0.673]. Both are 90/90 exact and accelerate zero of 30 prompts. The higher speed of same-GPU placement shows that placement and associated integration costs matter, but its continued loss rules out transfer as the sole explanation.

These controls are deliberately secondary. They qualify a slower path after the native fast attempt failed; they are not a rescued primary success. The verification microbenchmark explains lost batching opportunity, while the complete end-to-end controls include drafting and integration. We cannot recover a full R_{target} decomposition by substituting the forward-only ratio into Equation 2.

5.3 Arc B570: correctness-blocked diagnostics

Neither the fast path nor the serialized fallback supports a workload-global admissible optimization result. The latter's diagnostic speedup is 0.3087× [0.2927, 0.3246], with zero of 30 prompts faster. That interval describes timing on a partially mismatching path; its precision does not turn it into an admissible speed claim. The formal failure and stop rule remain the platform result. Board-energy observations likewise remain diagnostic and boundary-limited.

This negative control matters because it prevents an attractive simplification: selecting serialized verification is not by itself a sufficient correctness policy. Qualification must cover the actual path and intended workload. Exact subsets may be descriptively useful, but conditioning on those subsets after the run cannot establish global performance.

5.4 Core Ultra X7 358H: realized state changes the cost

The closed Phase-3 heterogeneous runner is exact on both CPU and NPU drafting: 90/90 pairs per path, with zero of 30 prompts faster. CPU achieves 0.734× [0.693, 0.771]; NPU achieves 0.352× [0.333, 0.369]. The direct round records show mean committed progress $C = 2.4631$ and 2.4584, respectively, approximately 2.46. This is complete committed output, not the accepted-draft counter used in older native records.

The target maintains persistent state, evaluates proposals, removes invalid candidate state when needed, and decodes correction or bonus output. Its complete measured region accounts for 87.34% of CPU-path speculative time. The NPU path instead spends 58.35% in the synchronous OpenVINO call that replays the full committed prefix and generates a proposal. Replay and proposal are inseparable within this measurement; we do not invent a replay-only value. Residual orchestration occupies only 0.04% and 0.02%, so it cannot explain the main loss.

Figure 4 separates isolated steady-state throughput from this realized request decomposition. The isolated NPU draft can outperform the CPU at the tested contexts, but this advantage does not survive the qualified state interface. The conclusion concerns that implementation, not insufficient NPU compute capability. A state-persistent intervention has not been implemented or measured.

Using arithmetic totals at a common boundary reconstructs Equation 2. CPU has $(D_{\mathrm{real}}, R_{\mathrm{target}}, X) = (0.4304, 2.9767, 0.00124)$ and aggregate speedup 0.7227×; NPU has $(4.1251, 2.9435, 0.00130)$ and 0.3477×. These agree with the ratios of total baseline/speculative times. They intentionally differ from the primary prompt-level geometric means in Table 3.

The earlier Phase-2 screen combined isolated drafting and a target-forward proxy under favorable residual-cost assumptions. It was an eligibility screen, not a validated end-to-end prediction. Phase 3 shows why those proxies must be replaced with complete target-side work and realized stateful draft cost. No end-to-end Phase-3 energy result is claimed, and source reconstruction retains its `STRONGLY_SUPPORTED` qualification.

5.5 Stable hardware-study conclusion

The tested systems expose four useful regimes: profitable and exact, exact but unprofitable, physically cheap but correctness-inadmissible, and state-interface-bound. Figure 5 maps them categorically; regimes can overlap and are not hardware ranks. Correctness determines which measured paths can support an optimization claim, then realized costs determine whether their committed progress pays. This conclusion requires neither a successful future prototype nor extrapolation from one workload to an entire device family.

6 Runtime Co-Design from the Measured Bottlenecks

Internal v0.1: proposed design only; implementation and measured results pending. This section reserves approximately 1.5–2 pages of the ten-page planning budget for a possible future measured contribution. It contains requirements, hypotheses, an interface proposal, and evaluation questions only. The hardware study is complete without it. No speed, energy, correctness, implementation, or ablation result is attributed to the proposed design.

6.1 Requirements traced to closed measurements

The direct stateful measurements motivate a narrow intervention: retain usable draft state across speculative rounds and reconcile it with the committed target history. The NPU replay-plus-proposal region motivates this requirement; the tiny measured residual does not motivate a broad rewrite of the orchestrator. The CPU result additionally warns that reducing draft cost alone may leave complete target-side work dominant. State persistence is therefore a hypothesis about a measured component, not a forecast of overall profitability.

The verification controls impose a separate requirement. A runtime must qualify the target path under its chosen output contract before using that path's timings to select an execution policy. A cheaper blocked path is not a candidate optimization under the current contract. B570 further requires that serialization be recorded as an execution setting, with exactness independently checked across the intended workload. A small probe cannot certify an unrestricted path selection policy.

The M4 contrast motivates an optional profitability guard based on realized quantities. Its purpose would be to decide whether a qualified speculative path is useful for the current workload, using direct committed progress and complete costs. It must not infer realized draft time from an isolated throughput number or count accepted draft tokens as all output progress. The proposed minimum study holds proposal width and device placement fixed; adaptive width and heterogeneous device selection would add separate questions only if later evidence justifies them.

6.2 Minimal state interface proposal

We propose a request-scoped draft-state object associated with a committed token prefix and a model/runtime configuration. A begin operation would initialize state from the prompt. A checkpoint operation would identify the com-

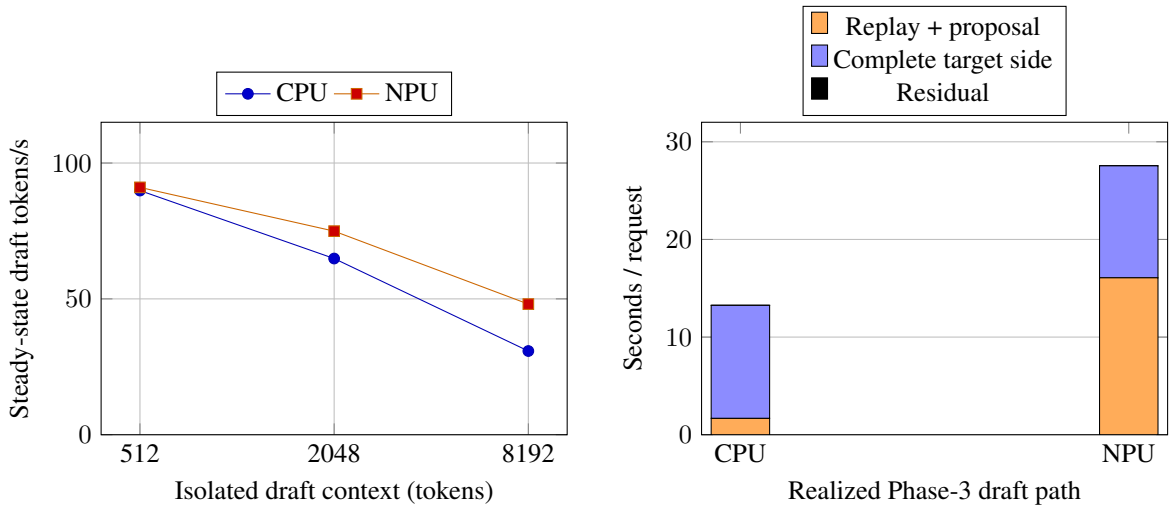


Figure 4. Why does isolated NPU speed not survive integration on 358H? Left: median isolated steady-state draft throughput (five repetitions/cell), excluding the first-token cost. Right: arithmetic mean complete post-prefill request time over 90 observations/path. CPU time is target-side dominated (87.34%); NPU time is replay-plus-proposal dominated (58.35%). Residuals are only 0.04%/0.02% and visually negligible. Separate boundaries preclude interpreting left-panel throughput as realized draft cost.

Profitable + exact
M4 HumanEval: primary

Exact but unprofitable
M4 24Q; 358H CPU/NPU: primary
RTX serialized controls: secondary

Physically cheap, inadmissible
RTX batched; B570 fast: diagnostic

State-interface-bound
358H NPU: measured mechanism

Figure 5. Which regimes survive the evidence distinctions? Categories summarize mechanisms, not an ordering of devices. NPU also belongs to the exact-negative regime. B570 serialized remains correctness-blocked diagnostic evidence, despite its slower execution.

mitted boundary before proposing. A propose operation would return candidate token IDs and a state handle describing their relation to that boundary. A reconcile operation would retain only the accepted draft prefix, incorporate any target correction or bonus token, and discard rejected candidate state. These are proposed interface responsibilities, not available functions in the measured adapter.

The target reference and its correctness contract would stay explicit. After every round, the target logical history, draft logical history, and committed output ledger would have to agree. Candidate state cannot become committed state merely because a proposal call completed. At request termination, all handles must be invalidated so a later request cannot inherit another prefix. If the backend cannot expose safe truncation or restoration, a replay path may be necessary; any such fallback cost must stay inside the realized draft boundary.

The interface should expose a complete draft-call timer, complete target-round timer, residual interval, and committed-token count under one enclosing decode window. Checkpoint creation, state copying, cache allocation, synchronization, and recovery must be charged to the responsible measured region. Merely moving work outside a narrow proposal timer would not establish an improvement. If replay and proposal remain inseparable, the evaluation should keep the aggregate region used by the hardware study rather than claiming an unavailable subcomponent decomposition.

The proposal does not require a new speculative algorithm, a new kernel, or a new learned draft. Its central test is whether the qualified backend can preserve useful state without changing token semantics or paying comparable reconciliation costs. Memory capacity and state-handle lifetime are therefore evaluation questions alongside latency. Persistent state and KV reuse have substantial prior art; their existence is not the intended novelty claim.

6.3 Correctness-qualified path selection

A minimal path registry would associate each target execution setting with its qualification scope: model artifacts,

runtime revision, numerical options, context/output conditions, and the evaluated workload. Selection would consider only paths admitted by the active contract. Changing the runtime, quantization, or execution shape would invalidate the relevant qualification rather than inherit it silently. This registry is proposed metadata and policy, not a claim that a finite test suite proves correctness for every possible input.

The evaluation must retain mismatches and blocked candidates. A path that fails remains a diagnostic observation with its failure scope visible. Recovery to target-only execution must preserve the same committed history and have its cost included. A future study could consider a different approximate-quality contract, but it would need its own quality metric and protocol; it cannot retrospectively relabel the frozen failures in this manuscript.

6.4 Planned evaluation questions and decision rule

A separate protocol should be frozen before observing intervention results. It should compare the current replay adapter, the proposed persistent-state adapter, and matched target-only decoding on the same prospective workload and timing boundary. Correctness qualification should exercise accepted prefixes, rejection, correction, bonus output, request reset, and output-budget boundaries. Full-workload exactness must accompany any promoted speed result; passing an initial gate alone is insufficient.

Two additional questions concern the implementation boundary. Does retaining state preserve its benefit as the committed prefix grows, or does checkpoint memory and reconciliation work simply replace replay? Does the same interface handle complete acceptance and early rejection without hidden full-prefix rebuilds? The proposed evaluation would record these events and their charged time. Results could then distinguish a reduction in executed work from a change in where the timer starts. Any context or event breakdown would be descriptive unless specified prospectively as a confirmatory comparison.

A paired prospective comparison should also keep reference configuration, output contract, initial prefill exclusion, and stopping behavior fixed. Run order should be randomized under the new protocol, and the primary estimator should be chosen before seeing the new data. Failures, fallback events, and memory pressure should remain in the record. A reduced cost on only a selected subset would not establish a workload-wide improvement. These requirements follow from the measured path and state boundaries; they do not assert that a proposed backend API can already meet them.

The key questions are whether the interface preserves token equivalence; whether realized draft cost decreases after all state-management work is counted; whether complete target-side cost changes; and whether the end-to-end prompt-level comparison improves. Component measurements and total-time accounting should reconcile. An energy claim would require a newly qualified end-to-end sensor boundary, which the closed 358H campaign does not supply.

If state persistence reduces draft cost but leaves the system unprofitable, the result should be reported in that form. If correctness fails, the timing remains diagnostic. If the implementation is unavailable or yields no defensible contribution, this entire section can be removed without changing the hardware-study conclusion. A successful future intervention would need measured evidence and a revised abstract, contribution list, and conclusion; none is assumed in manuscript v0.1.

7 Related Work

Performance and cost accounting. Speculative decoding amortizes target computation while preserving a target sampling distribution under its algorithmic assumptions (Leviathan et al., 2023). Liu et al. (2026) systematically study speculative methods in vLLM, including verification dominance and workload/batch effects. Kim et al. (2026) control shared runtime policies and express speedup through committed work and conditional execution cost. Lyu et al. (2026) derive profitability conditions for edge–cloud execution. We neither introduce performance characterization nor claim a novel cost/commit identity. Our focus is the measured cost of the paths remaining after a strict greedy-reference gate and the stateful interface actually used on consumer systems.

Heterogeneous execution. Dovetail places drafting on a GPU and target verification on a CPU (Zhang et al., 2025); DuoDecoding studies CPU/GPU placement with hardware-aware draft budgets (Lv et al., 2025). Mirror-SD explicitly considers GPU/NPU mapping and complementary speculative pipelines (Bhendawade et al., 2025). Heterogeneous speculative execution is therefore established prior art. Our negative heterogeneous outcome characterizes the qualified interface and does not contradict acceleration under other placements or runtimes.

Correctness and numerical reproducibility. Zhang et al. (2026) analyze batch speculative-decoding synchronization errors and distinguish them from residual numerical mismatches. Their current paper revises the earlier *Batch Speculative Decoding Done Right* preprint. Yuan et al. (2025) study numerical sources of output variation across system configurations; LLM-42 uses verified speculation to enforce deterministic inference (Gond et al., 2026). Our CUDA observation is not a new numerical phenomenon, and we do not equate it with ragged-batch synchronization bugs. We measure its consequence for an admissible target path, with

B570 showing why serialization alone cannot certify the intended workload.

State and memory reuse. Draft-side state management also has prior art. Yuan et al. (2026) maintain incrementally updated compressed draft KV memory for long contexts. Such designs differ from the full-prefix replay interface measured here, but preclude treating state reuse itself as a new idea. Our evidence identifies a realized replay-plus-proposal cost in a consumer OpenVINO NPU path. Any later co-design contribution must be established by its own measured behavior rather than by adopting persistence as an architectural label.

8 LIMITATIONS

This study covers four consumer systems and a narrow Qwen3 model family. Hardware, operating systems, runtime versions, backends, and some quantization choices co-vary. The results establish observed regimes and within-platform mechanisms, not an architectural ranking or a general deployment law. Serving concurrency, production load, and broader model families are outside the measured scope.

Token equivalence is deliberately strict and conditional on the frozen reference; a different quality contract could admit other paths. HumanEval functional execution and pass@k were not evaluated. Serialized B570 outputs still mismatch, and the CUDA evidence is bounded to the captured stack and case. No claim of general backend unsuitability follows.

Campaign-specific estimators and bootstrap implementations differ. Prompt intervals do not cover all system-state uncertainty, including M4 session/thermal effects. The M4 corrected synthetic-ceiling preregistration remains unexecuted. Decode timing excludes initial prefill and loading, so it is not whole-request user latency. Energy sensor domains differ; there is no global energy ranking or end-to-end 358H Phase-3 energy claim.

Windows raw per-run provenance is weaker than that of the other closed campaigns. The 358H draft-source reconstruction is `STRONGLY_SUPPORTED`, not proven historical upstream identity. Replay and proposal are measured together, preventing separate causal timing attribution to either. No measured runtime intervention exists in this version; persistent state, path selection, and profitability control remain proposals. These limitations bound the hardware conclusion independently of the future software decision.

9 CONCLUSION

Under the chosen strict greedy-reference contract, consumer speculative decoding spans profitable exact, exact-negative, correctness-inadmissible, and state-interface-bound regimes. The physically cheapest target-forward path may be excluded by correctness, and isolated accelerator throughput may fail to represent the stateful draft cost paid in the speculative loop. Retaining both M4 workloads, the blocked and serialized verification controls, and the complete 358H decomposition makes this a coherent hardware characterization without a new runtime result. The hardware-study conclusion is stable; the final paper conclusion remains provisional pending a separately measured software contribution or the decision to remove that reserved scope.

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